

Every Prompt Has a Price: A Toolbox for Tracking the Environmental Cost of GenAI Use

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Abstract

As generative AI becomes embedded across scientific, creative, and commercial workflows, its environmental costs—from electricity and water consumption to carbon emissions—remain largely hidden. While existing research has highlighted the footprint of model training, everyday GenAI use (inference) also consumes significant resources, this poster introduces the GenAI Environmental Impact Toolbox, a curated collection of tools designed to help researchers, developers, and practitioners estimate and reflect on the environmental cost of their GenAI activities. The toolbox includes browser-based calculators, integrated code libraries, and benchmark leaderboards, tailored to different technical levels and use cases. We analyze each tool’s design, scope, and usability, and outline opportunities for collective maintenance and future development. We advocate for integrating environmental awareness into human-AI interaction design through real-time feedback, energy labels, and hybrid intelligence workflows. The toolbox aims to make sustainability part of everyday GenAI use—empowering users to see and act on the true cost of every prompt.

CCS Concepts

• **Software and its Engineering**; • **Software organization and its properties**; • **Computing methodologies**; • **Machine learning**;

Keywords

GenAI, Environment, Carbon, Hybrid Intelligence, Toolkit

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1 Introduction

Generative AI (GenAI) has become increasingly integrated into research, development, and industry workflows. Researchers use it for tasks ranging from academic writing and prototyping to large-scale data analysis and modeling [1, 2]. At the same time, companies are embedding GenAI tools across product pipelines, automating processes, and accelerating innovation at scale [3, 4]. This widespread adoption, while transformative, carries a largely invisible environmental footprint [5, 6, 30].

AI systems—and the massive data infrastructure that powers them—consume vast quantities of electricity, fresh water, and rare minerals, most of which are difficult to recycle and contribute to growing electronic waste [10]. Hyperscale data centers, often exceeding one million square feet, are central to GenAI operations. In 2022, data centers consumed 460 terawatt-hours—nearly matching the energy use of entire countries such as France. By 2026, this figure is expected to more than double [13].

While the environmental cost of training large models is increasingly acknowledged, the everyday use of GenAI—inference—also drives significant energy demand. A single large model like GPT-3 emitted an estimated 552 tons of CO₂ during training [13], but the emissions from widespread, repeated usage by millions of users are substantial and ongoing. Moreover, most data centers still rely on fossil-fuel-powered energy grids or on-site diesel generators [8]. Water usage is another key factor: cooling a single large data center can require over one million liters of drinking water per day [12]. These centers are often located in already water-stressed regions,

where they compete directly with local communities and ecosystems. The chemicals used in data center cooling—such as PFAS and corrosion inhibitors—further pollute regional water cycles [14].

As GenAI becomes ubiquitous, we must ask not only what AI can do—but also what it costs. That cost is not just financial or technical—it’s environmental, and increasingly global. There are growing efforts to quantify and reduce the environmental footprint of AI. A comprehensive life cycle analysis (LCA) approach, such as the one proposed by the Coalition for Sustainable AI (2026), includes the following components: **Software impacts:** energy and compute demands from data collection, model development, training, deployment, and ongoing inference. **Hardware impacts:** the environmental toll of rare earth mineral extraction, manufacturing, transport, cooling, energy, and eventual e-waste disposal.

While international standards and policy frameworks are still evolving, researchers and practitioners can already take concrete steps to assess their GenAI use. Several tools have emerged to help estimate carbon and energy impacts during model training or usage—some embedded directly into development environments, others accessible through web-based calculators [9]. Since the tools vary widely in scope, usability and intended audience, and are often unknown and underused, we hope to shed some light on what is already available, and what is still missing.

2 Our contribution: The GenAI environmental impact toolbox

To support more informed and sustainable use of GenAI, we compiled the **GenAI Environmental Impact Toolbox**, a curated selection of 11 tools that estimate or track the energy, carbon, and, in some cases, water impacts of GenAI applications. The tools range from lightweight web calculators to integrated code libraries and large-scale model benchmarks. Each offers different levels of complexity, precision, and applicability depending on the user’s needs, expertise, and use case (see Figure 2).

The toolbox includes browser-based calculators designed for accessibility and ease of use. For example, EcoLogITs [24] allows users to input model parameters, select a region, and receive visualized estimates of electricity consumption, CO₂ emissions, water use, and other impacts. It also translates these into intuitive equivalents—such as kilometers driven or plastic bottles produced—helping to bridge the gap between abstract energy data and tangible outcomes. Similarly, Green Algorithms provides location-sensitive calculations for carbon intensity and allows researchers to adjust

hardware and runtime variables to refine their estimates [21]. These tools require no programming knowledge and are well-suited to exploratory use by scientists, educators, and policy-facing audiences.

For more technical workflows, Python libraries such as CodeCarbon [15], Carbontracker [17], and ECO2AI [18] can be embedded directly into machine learning pipelines. These tools monitor the resource usage of CPUs and GPUs in real time and estimate associated emissions using regional energy profiles. CodeCarbon, for instance, can be integrated with minimal setup into PyTorch or TensorFlow scripts and provides detailed tracking of emissions over time, along with contextual analogies to help interpret impact. Carbontracker and ECO2AI add support for hardware-level measurements and local log storage, allowing researchers to track emissions across experiments and infrastructure. These tools are most useful for developers, scientists, and technical teams running fine-tuning or training workloads.

Several leaderboard-style tools, including Hugging Face’s AI Energy Score [19] and the ML.ENERGY Leaderboard [20], offer comparative benchmarks of energy efficiency across models and tasks. These tools run standardized tests using shared hardware configurations, allowing users to select models not only based on performance but also on energy consumption and environmental impact. Though these tools focus primarily on inference, they help make efficiency a first-class metric in model selection—especially for researchers deploying GenAI at scale or across multiple workflows.

In addition, the toolbox includes more specialized and domain-specific tools. The HCI GenAI CO₂ST Calculator [22], for example, was designed specifically for researchers working in Human-Computer Interaction and offers estimates of emissions based on use patterns drawn from published research. Green Algorithms [16] for HPC extends sustainability tracking to high-performance computing environments, enabling users to calculate the energy and emissions associated with compute-intensive tasks using log data from SLURM workload managers. These tools reflect emerging interest in tailoring sustainability practices to specific disciplines and infrastructures. See Figure 1 for screenshots of examples of tools.

Across these tools, we observed a number of **common tensions** which we include in the toolbox as *weaknesses*. Some tools prioritize accessibility over precision, relying on generalized assumptions or static inputs, while others demand high technical fluency to produce more accurate results. Most tools remain focused on carbon emissions from inference impact, neglecting the



Figure 1: Examples of three kinds of tools from the toolkit: ML CO₂ impact calculator, ECO2AI python library, and AI Energy Score leaderboard

Title (year)	Author/ Publisher	Category	Who is it for	What's evaluated	Scope	How it works
Zeus Project (2025, regular updates)	Jae-Won Chung / Zeus Team	Code	Developers	Energy Consumption, Training Impact	Training Impact	Instantiates PowerMonitor to poll the device's power API, collecting deduplicated power samples in memory during workload execution.
Code Carbon	Code Carbon Contributors	Code	Developers	Carbon Emissions	Training impact	Code is downloaded and embedded.
Green Algorithms - 4 HPC	Green Algorithms	Code	Developers	Carbon Footprint, Energy use	Training & Inference/serving impact	Pulls usage stats from HPC workload manager (SLURM) logs to estimate carbon footprint. Reports energy use, emissions, compute/memory efficiency, and failed job impact. Adaptable to other workload managers.
Carbontracker	Peter Roffes	Code	Developers	Carbon Emissions, Energy Consumption, Runtime Performance	Training & Inference/serving impact	Users submit ML code with parameters; server runs tasks, tracks resource use, and calculates emissions. Results are shown in a dashboard.
ECO2AI	Budennyi et al	Code	Developers	Carbon Emissions	Training & Inference/serving impact	Python library that tracks CPU/GPU energy use and estimates CO ₂ emissions using regional coefficients. Add a few lines to your script; results are logged locally with session metadata.
AI Energy Score Leaderboard / Project (2025)	Luccioni et. al. (Hugging Face)	Leaderboard	Researchers, General audiences	Energy Efficiency	Inference/serving impact	Runs 1,000-task benchmarks sampled from real-world and standard datasets. Models tested on NVIDIA H100 GPUs; energy use tracked with CodeCarbon and launched via Neuralwatt benchmarks.
MLENERGY Leaderboard (current version: 2025)	MLENERGY Initiative team	Leaderboard	Researchers, General audiences	Energy Efficiency	Inference/serving impact	Benchmarks LLMs, MLLMs, and diffusion models using vLLM/xDIT on H100 and B200 GPUs. Measures GPU energy via Zeus after thermal stabilization. Sweeps configurations to map time-energy tradeoffs and suggest optimal setups.
Green Algorithms - Carbon footprint calculator	Green Algorithms	Webbased calculator	Researchers, General audiences	Carbon Emissions	Inference/serving impact	Calculates carbon footprint by multiplying estimated energy use with regional carbon intensity. Inputs include runtime, core/memory specs, usage, and data center PUE. Adjusts for repeated runs and offers location-based insights.
HCI GenAI CO2ST Calculator	Nanna Inie, Jeanette Falk, Raghavendra Selvan	Webbased calculator	Researchers, General audiences	Carbon Emissions	Inference/serving impact	Users input research phase, AI use type, model, and document count. Calculator estimates carbon footprint using prior local inference tests and Carbontracker data, based on tasks from CHI 2024 HCI research.
Machine Learning (ML) CO2 Impact Calculator	Mila (Quebec AI Institute) and Element AI (now ServiceNow)	Webbased calculator	Researchers, General audiences	Carbon Emissions	Inference/serving impact	Users select hardware, runtime, and cloud provider. Tool estimates raw and offset carbon emissions by multiplying energy use with region-specific carbon intensity values.
EcoLogITs Calculator	GenAI Impact non-profit	Webinterface tool, Calculator	Researchers, General audiences	Electricity consumption, GHG Emissions, Abiotic Resources, Primary Energy, Water	Inference/serving impact	Estimates energy and resource use based on user inputs like prompt, model, hardware, and assumed data center location. Outputs include electricity, GHGs, water, and more.

Figure 2: Screenshot of selected columns from the GenAI Environmental Impact Evaluation Toolbox

environmental costs of data collection, hardware manufacturing, or periods of model training. Some tools convey environmental costs less intuitively than others, presenting only abstract measures which can complicate reflection and decision-making, particularly among non-expert users.

By gathering these tools into a cohesive and openly available resource, the GenAI Environmental Impact Toolbox lowers the barrier for responsible use of generative AI across technical and non-technical domains. A full overview, including tool functions, intended audiences, and methodological notes and strengths and weaknesses, is available in our open-access spreadsheet at: <http://h-infinity.com/resources/genai-toolbox>

3 Discussion: Towards Meaningful Use and Evolution of GenAI Environmental Tools

The tools presented in this toolbox offer a practical starting point for researchers, practitioners, and everyday GenAI users to better understand and mitigate the environmental impacts of their work. Although these tools differ in complexity, audience, and functionality—from simple web calculators to integrated Python libraries—they all play a role in illuminating the often-invisible environmental cost of GenAI.

In practice, users can select tools that align with their technical experience and typical GenAI workflows. For instance, practitioners working with commercial AI APIs may begin with lightweight tools such as the EcoLogITs Calculator [24], which provides quick,

code-free estimates of environmental impact based on inputs like model, provider, and prompt. Developers and researchers training or fine-tuning models can integrate libraries like CodeCarbon [15] or Carbontracker [17] into their Python environments to track emissions in real time. Those involved in model evaluation can use tools such as the AI Energy Scoreboard [19], which supports comparative selection of models based on energy efficiency metrics. As AI continues to move from labs into general-purpose enterprise and creative applications, embedding these tools into design pipelines, development environments, and platform APIs could help normalize sustainable choices and support energy-aware development by default.

To support the **long-term growth and relevance of the toolbox**, we propose a community-maintained, open-source platform that enables wiki-style contributions. Such a platform would allow users and developers to submit new tools, visualizations, and benchmarks, while also encouraging researchers to contribute domain-specific case studies or usage scenarios. Tool developers could share integration examples, usability updates, and performance metrics to support collective learning. Inspired by collaborative ecosystems like Papers with Code and Hugging Face, this model would lower entry barriers and help keep the toolbox current, diverse, and widely accessible.

Our analysis of the tools' design and user experience revealed several **common challenges and areas for improvement**. First, there is significant variation in accessibility and comprehension: some tools translate emissions data into familiar analogies (like kilometers driven or hours of TV watched), while others assume prior technical knowledge. Ensuring that impact information is easy to interpret—even for non-expert users—is essential for wider adoption. Second, the level of interaction granularity matters. Tools that provide real-time feedback during model training or prompting enable more informed decisions and foster reflective use. In contrast, static calculators—though useful—may feel disconnected from real-world development patterns. Third, integration potential remains limited. Few tools offer APIs, plug-ins, or seamless connections to the environments where GenAI is most often used, such as Jupyter notebooks, IDEs, or low-code platforms. Lastly, the toolbox highlights that the range of environmental impacts measured is often very narrow and most tools focus only on carbon emissions and energy efficiency. Other environmental impacts like water-usage or abiotic resource consumption remain largely underrepresented.

To address these gaps, we advocate for human-centered design principles in future development of environmental impact tools. These should include real-time, contextual feedback; tiered information depth for novice and expert users; privacy-respecting defaults; minimal setup friction; and visual or comparative cues that help users make more sustainable infrastructure and model choices. We further call for stronger transparency from all involved stakeholders, to enable more encompassing measuring of environmental impacts of GenAI across the board.

Looking ahead, AI environmental labels represent a promising direction [29]. These would provide standardized, easily understood summaries of a model or system's expected energy use, carbon emissions, and water consumption per task or prompt. Like nutrition

labels, these could be embedded into model cards, platform interfaces, or documentation—helping users make informed decisions at the point of interaction.

Another strategy is to rethink the way humans and AI systems collaborate. Hybrid Intelligence [25–27] approaches encourage thoughtful, goal-oriented interaction with GenAI, moving away from brute-force prompting or trial-and-error experimentation [27]. This has been proposed by the International Telecommunication Union to reduce redundant or inefficient queries, and designing for targeted, high-leverage interactions, can help minimize computational waste [28]. For example, future GenAI systems could offer “impact-conscious modes” or “low-energy query suggestions” that prioritize efficiency without sacrificing user experience. Interfaces could also guide users to refine their goals, co-construct effective prompts, or draw from previously successful interactions—all of which reduce unnecessary system load and environmental cost.

Together, these design, community, and interaction strategies chart a path toward more responsible and sustainable GenAI use—helping users not only see the environmental cost of their prompts but meaningfully act on it.

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References

- [1] Sébastien Bubeck, Varun Chandrasekaran, Ronitt Rubinfeld, and Yi Zhang. 2023. *Sparks of Artificial General Intelligence: Early Experiments with GPT-4*. arXiv: 2303.12712. <https://arxiv.org/abs/2303.12712>
- [2] Minsuk Lee, Adam D. Hockenberry, Jacob Andreas, and Yonatan Bisk. 2023. *Evaluating Human-Centered Use Cases for Large Language Models*. In Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23). ACM, New York, NY, USA. <https://doi.org/10.1145/3544548.3580657>
- [3] James Manyika, Gary Bradski, Jack Clark, and Eric Schmidt. 2023. *The State of AI in 2023*. McKinsey Global Institute. <https://www.mckinsey.com/mgi/overview/in-the-news/the-state-of-ai-in-2023>
- [4] OpenAI. 2024. *GPTs in the Workplace: Usage Trends and Enterprise Applications*. OpenAI Research Blog. <https://openai.com/research/gpts-in-the-workplace>
- [5] Emma Strubell, Ananya Ganesh, and Andrew McCallum. 2019. Energy and Policy Considerations for Deep Learning in NLP. In Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL 2019), 3645–3650. <https://doi.org/10.18653/v1/P19-1355>
- [6] Alexandra Sasha Luccioni, Sylvain Viguier, and Yoshua Bengio. 2022. *Estimating the Carbon Footprint of BLOOM, a 176B Parameter Language Model*. arXiv:2211.02001. <https://arxiv.org/abs/2211.02001>
- [7] Center of Scientific Evidence in Public Issues. 2025. Data Centers and Environmental Considerations. Retrieved from <https://www.aaas.org/sites/default/files/2025-09/Data%20Centers%20Fact%20Sheet%202%20-%20Data%20Centers%20and%20Environmental%20Considerations.pdf>
- [8] Amber X. Chen. 2025. A.I. Is on the Rise, and So Is the Environmental Impact of the Data Centers That Drive It. [www.smithsonianmag.com](https://www.smithsonianmag.com/science-nature/with-ai-on-the-rise-what-will-be-the-environmental-impacts-of-data-centers-180987379/). Retrieved from <https://www.smithsonianmag.com/science-nature/with-ai-on-the-rise-what-will-be-the-environmental-impacts-of-data-centers-180987379/>
- [9] Coalition for Sustainable AI. 2026. A shared vision for the Coalition for Sustainable AI. Retrieved from <https://www.sustainablecoalition.org/coalition/>
- [10] Ministère de l'Écologie. 2025. Standardization for AI Environmental Sustainability. Retrieved from https://www.sustainablecoalition.org/wp-content/uploads/Standardization_AI_Sustainability.pdf
- [11] Mary Romano, Katherine Logue, Molly Tipping, April McLane, Richard Hunnewell, Devanashi Kaura, and Rotem Kanny. 2025. The Resource-Draining Origins of Our AI Assistants: Potential Policy Solutions to the Unregulated Growth of Georgia's Data Center Industry. *Journal of Science Policy & Governance* 26, 1. <https://doi.org/10.38126/JSPG260110>
- [12] Ken Silverstein. 2026. America's AI Boom Is Running Into An Unplanned Water Problem. Retrieved from <https://www.forbes.com/sites/kensilverstein/2026/01/>

- 11/americas-ai-boom-is-running-into-an-unplanned-water-problem/
- [13] Adam Zewe. 2025. Explained: Generative AI's environmental impact. Retrieved from <https://news.mit.edu/2025/explained-generative-ai-environmental-impact-0117>
- [14] Zeus Team. 2025. *Zeus: Energy Measurement and Optimization for Deep Learning Workloads*. Retrieved January 23, 2026 from <https://ml.energy/zeus/>
- [15] CodeCarbon Contributors. 2025. *CodeCarbon: Track and Reduce Your ML Emissions*. Retrieved January 23, 2026 from <https://codecarbon.io/>
- [16] Green Algorithms. 2025. *Green Algorithms for HPC: Estimate Carbon Footprint of High Performance Computing*. Retrieved January 23, 2026 from <https://www.green-algorithms.org/GA4HPC/>
- [17] Rolfes, Peter. 2025. *Carbontracker: A Python Package to Track Carbon Emissions of Machine Learning Workloads*. Retrieved January 23, 2026 from <https://pypi.org/project/carbontracking/>
- [18] Budennyy, Sergey A., Vladimir D. Lazarev, Nikolay N. Zakharenko, Alexander N. Korovin, Olga A. Plosskaya, Dmitry V. Dimitrov, Vladimir S. Akhrikin, Ivan V. Pavlov, Ivan V. Oseledets, Ivan S. Barsola, Ivan V. Egorov, Anna A. Kosterina, and Leonid E. Zhukov. 2023. eco2AI: Carbon Emissions Tracking of Machine Learning Models as the First Step Towards Sustainable AI. *Doklady Mathematics*. <https://doi.org/10.1134/S1064562422060230>
- [19] Luccioni, Alexandra Sasha, et al. 2025. *AI Energy Score Leaderboard*. Hugging Face. Retrieved January 23, 2026 from <https://huggingface.co/spaces/AIEnergyScore/Leaderboard>
- [20] MLENERGY Initiative. 2025. *MLENERGY Leaderboard: Time–Energy Tradeoffs for Generative AI*. Retrieved January 23, 2026 from <https://ml.energy/leaderboard/>
- [21] Green Algorithms. 2025. *Green Algorithms Carbon Footprint Calculator*. Retrieved January 23, 2026 from <https://calculator.green-algorithms.org/>
- [22] Inie, Nanna, Jeanette Falk, and Raghavendra Selvan. 2025. *HCI GenAI CO2ST Calculator*. Retrieved January 23, 2026 from <https://www.hcico2st.com/>
- [23] Mila and Element AI. 2025. *Machine Learning CO2 Impact Calculator*. Retrieved January 23, 2026 from <https://mlco2.github.io/impact/#learn>
- [24] GenAI Impact. 2025. *EcoLogITs Calculator: Estimate Environmental Impacts of GenAI Use*. Hugging Face. Retrieved January 23, 2026 from <https://huggingface.co/spaces/genai-impact/ecologits-calculator>
- [25] Zeynep Akata, Dan Balliet, Maarten De Rijke, Frank Dignum, Virginia Dignum, Gusztai Eiben, Antske Fokkens, Davide Grossi, Koen Hindriks, Holger Hoos, et al. 2020. A research agenda for hybrid intelligence: augmenting human intellect with collaborative, adaptive, responsible, and explainable artificial intelligence. *Computer* 53, 8 (2020), 18–28.
- [26] Yaoli Mao, Janet Rafner, Yi Wang, and Jacob Sherson. 2023. A hybrid intelligence approach to training generative design assistants: partnership between human experts and AI enhanced co-creative tools. In *HHAI 2023: Augmenting Human Intellect*. IOS Press, 108–123.
- [27] Jacob Sherson, Janet Rafner, and Safi Büyükgüzel. 2024. Operational criteria of hybrid intelligence for generative AI virtual assistants. In *HHAI 2024: Hybrid Human AI Systems for the Social Good*. IOS Press, 475–477.
- [28] International Telecommunication Union (ITU). 2025. *Measuring what matters: How to assess AI's environmental impact (S-GEN-GDA.001-2025)*. Retrieved January 23, 2026 from <https://www.itu.int/hub/publication/s-gen-gda-001-2025/>
- [29] van der Staay, Anne, Robert Fischer, Martin Wischniewski, Katrin Poitz, and Christian Janiesch. 2025. Reflective Design Theorizing with User Interviews: A Case Study for AI Energy Labels. In *Proceedings of the International Conference on Design Science Research in Information Systems and Technology (DESRIST 2025)*, 66–80. Springer Nature Switzerland, Cham.
- [30] Utz, V. and DiPaola, S. 2023. *Climate Implications of Diffusion-based Generative Visual AI Systems and their Mass Adoption*. In *Proceedings of the 14th International Conference on Computational Creativity (ICCC '23)*, 264–272.